Enhancement Three Narrative: Databases

The artifact for this category is the same animal shelter dashboard from CS 340 that I used for my algorithms and data structures enhancement. Because this one project covers two categories, the code files are the same ones from that milestone, but this narrative focuses only on the database side. The dashboard connects to a MongoDB database of animal shelter records through a Python CRUD module called CRUD_Python_Module.py, which handles creating, reading, updating, and deleting records.

I chose to enhance the database side of this artifact because it had a few real problems that would matter in a production app. The biggest one was security. The original notebook had the database username and password typed directly into the code. Anyone who could see the notebook, including anyone viewing my public GitHub repository, could see the credentials. I changed the CRUD module so it reads the username and password from environment variables called AAC_USER and AAC_PASS, and raises a clear error if they are not set. The notebook no longer contains any credentials at all.

The second change was adding an index. Every rescue filter on the dashboard searches by animal_type and breed, and without an index MongoDB has to scan every record in the collection to find matches. I added a compound index on those two fields when the CRUD module connects, so those searches can go straight to the matching records. The create_index call does nothing if the index already exists, so it is safe to run every time the app starts.

The third change was a safeguard on the update and delete methods. Both used MongoDB's update_many and delete_many, and in MongoDB an empty query matches every document in the collection. That means one bad call with an empty query could change or delete every record in the database. I added a check to both methods so they refuse to run if the query is empty and return zero instead.

These changes mainly support the course outcome on developing a security mindset. Moving credentials out of the code and blocking accidental mass updates and deletes are both about anticipating how things could go wrong and closing those gaps before they cause damage.

The index supports the outcome on designing and evaluating computing solutions, since it is a tradeoff between a little extra storage and much faster searches on the fields the app uses most.

One thing I learned while doing this is that removing a password from the code does not remove it from the project's history. My original password is still visible in the earlier commits on GitHub. In a real job, the right fix would be to change that password so the old one no longer works. That was an eye opener for me about how easy it is to leak sensitive information through version control. The main challenge was keeping the changes focused, since it would have been easy to keep adding more checks. I stuck to the three changes in my enhancement plan so each one was easy to explain and test.